Atlantic halibut (*Hippoglossus hippoglossus*)

Summary Scores

- **Sensitivity:** High
- **Exposure:** Moderate
- **Vulnerability:** Moderate

Sensitivity

Sensitivity Attributes	Score	LH DQ	Tally
Adult Mobility	1.2	2.53	
Complexity in Reproductive Strategy	1.96	2.6	
Dispersal of Early Life Stages	1.2	2.9	
Egg and Larval Survival and Settlement Requirements	1.92	2.05	
Historical Variability in Temperature	2.16	3	
Juvenile Habitat Requirements	1.2	3	
Other Stressors	1.16	2.05	
Population Growth Rate	3.56	2.74	
Prey Specificity	1.2	3	
Relationship to Calcified Organisms (due to impacts from Ocean Acidification)	1.2	2.8	
Spawning Cycle	2.64	2.7	
Stock Size/Status	3.76	1.9	
Total Sensitivity | Certainty	**High (3) | 1**		

Exposure

Exposure Factors	Mean	SD	Tally
Bottom Temperature	2.17	1.03	
Bottom Oxygen	2.79	1.00	
Bottom Salinity	2.34	1.07	
Bottom Aragonite Solubility	1.23	0.57	
Small zooplankton integrated biomass	1.24	0.25	
Medium zooplankton integrated biomass	1.32	0.27	
Large zooplankton integrated biomass	1.36	0.37	
Total Exposure	**Moderate (1.82)**	**0.90**	

Vulnerability

Vulnerability	Mean	SD	Tally
Total Vulnerability	**Moderate (5.47)**	**2.7**	

Low Moderate High Very High

Figure 1: Summary Sensitivity, Exposure, and Vulnerability Scores and Score Distributions for Atlantic halibut. Exposure and Vulnerability scores are averaged spatially across the species distribution and within stocks if applicable (see Figure 5 and Figure 4). Life History Data Quality (LH DQ) and Standard Deviations (SD) are reported for senstivity and exposure/vulnerability scores, respectively.

Changes from Previous Assessment(s)

Table 1: Changes to Vulnerability, Exposure, and Sensitivity scores relative to Previous Assessments. If a citation for the previous assessment is not provided, it comes from Hare et al. (2016).

Metric	Previous Assessment(s)	Current Assessment	Change
Sensitivity	High	High	None
Exposure	High	Moderate	Decrease
Vulnerability	High	Moderate	Decrease
Directional Effect	Negative	Negative	None

Stocks

Atlantic halibut is managed as a single stock or stock information was not considered for this iteration of the CVA.

Score Descriptions:

Sensitivity:

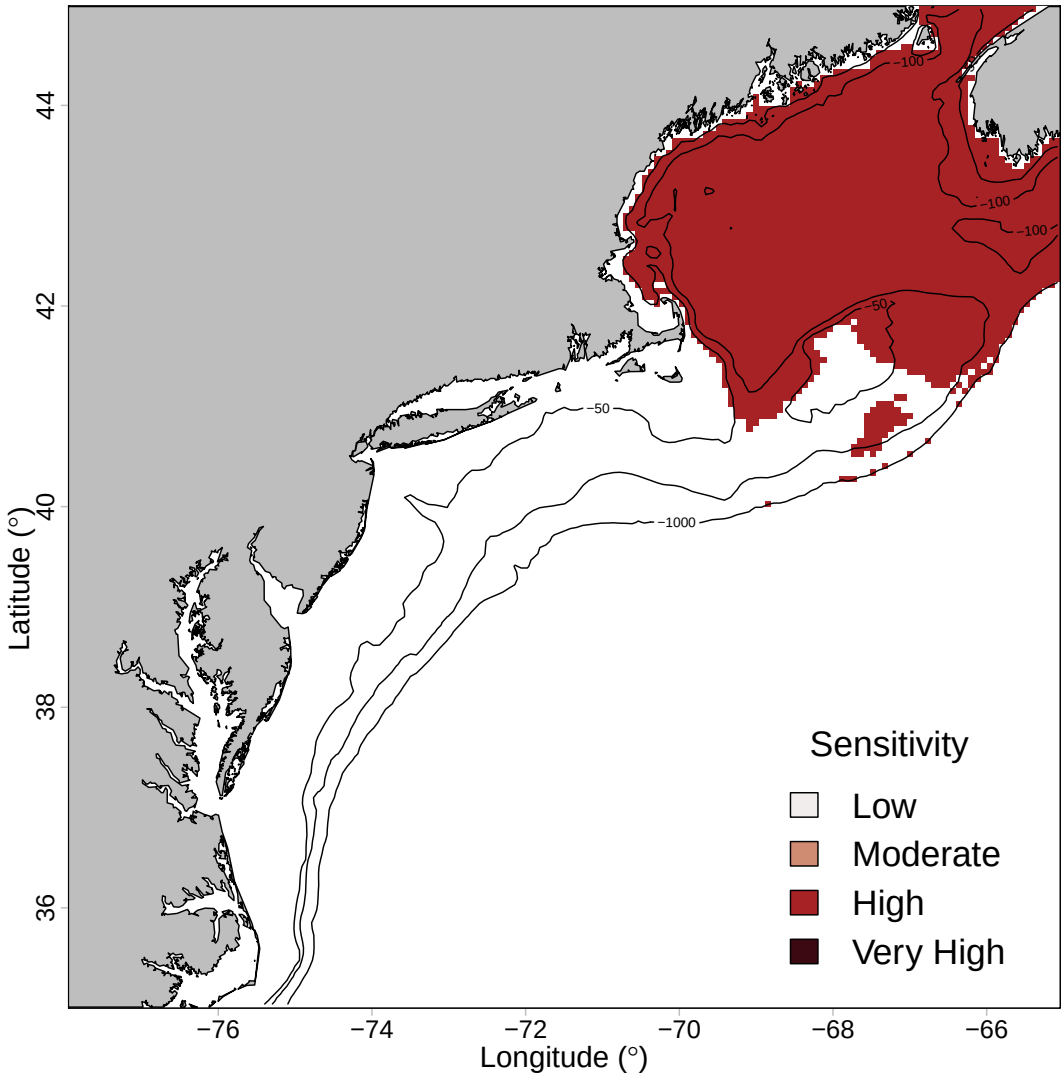


Figure 2: Atlantic halibut Sensitivity on the Northeast Shelf

Atlantic halibut scored **High** with a certainty of 100%. The score was driven by Stock Size (3.8) and Population Growth Rate (3.6) attributes. While it did not contribute to the score, it is worth noting that the only other attribute that scored higher than 2.5 was Spawning Cycle (2.6) (Figure 1).

Halibut mature slowly and are long lived species, living upwards of 50 years and maturing between 5-13 years (Armsworthy, Trzcinski, and Campana 2014).

Halibut also received a score of high sensitivity in the previous Climate Vulnerability Assessment. This score was also driven by Stock Size/Status and Population Growth Rate (Hare et al. 2016).

Exposure:

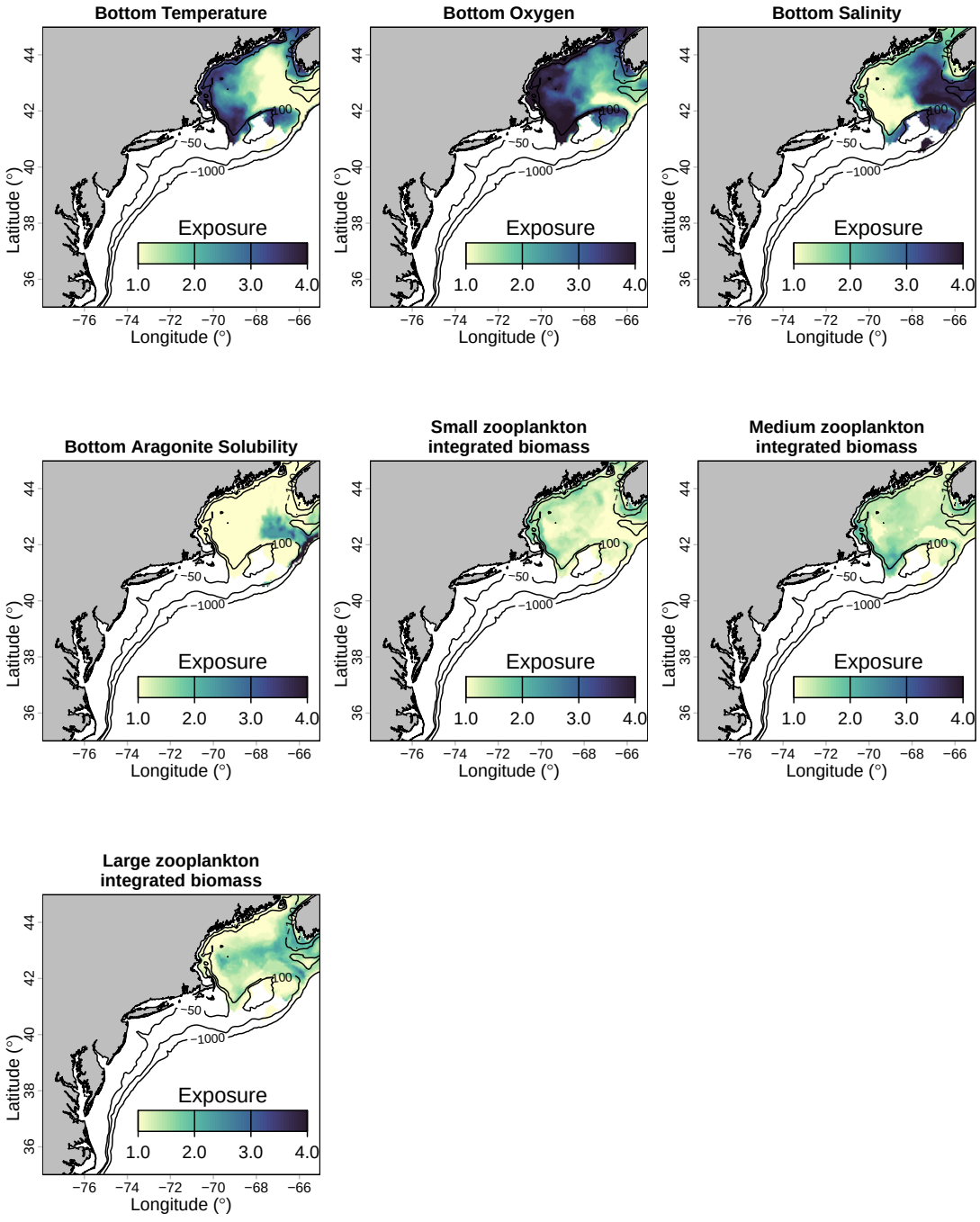


Figure 3: Maps of each Exposure Factor for Atlantic halibut. Exposure was calculated between 2014-2023 and 2025-2035

Atlantic halibut received a **Moderate** mean Exposure score (1.82 +/- 0.90) (Figure 1). Exposure scores were high along the coast of the Gulf of Maine and over Georges Bank, driven by high exposure to bottom

temperature, oxygen and salinity changes (Figure 4, Figure 3). However, Low exposures over the middle of the Gulf of Maine drove mean scores down.

In the previous Assessment, Atlantic halibut received a High exposure score, driven by sea surface temperatures and ocean acidification (Hare et al. 2016).

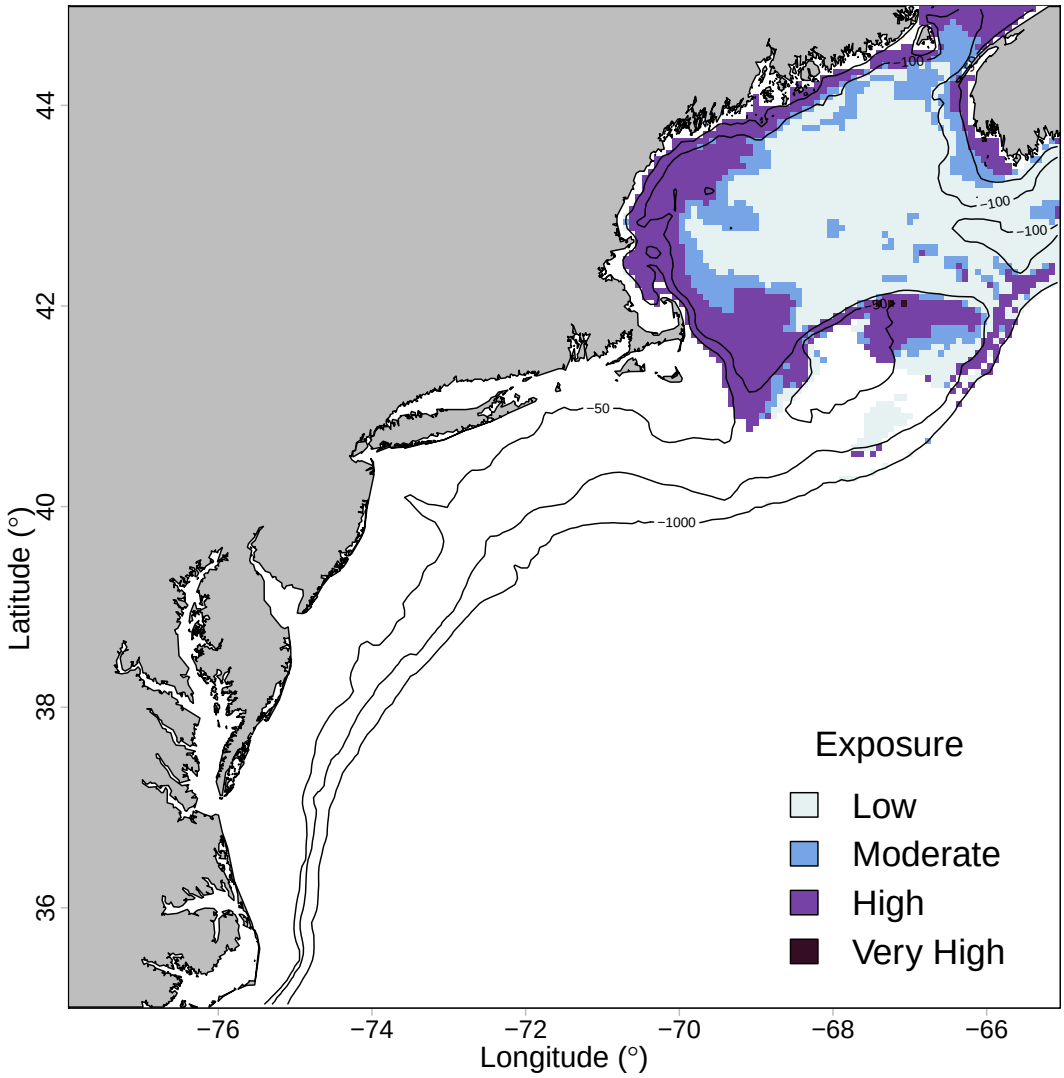


Figure 4: Total Exposure on the Northeast Shelf for Atlantic halibut. Total Exposure is calculated by applying the logic rule from Hare et al. (2016) within grid cells across exposure factors (see Figure 3)

Vulnerability:

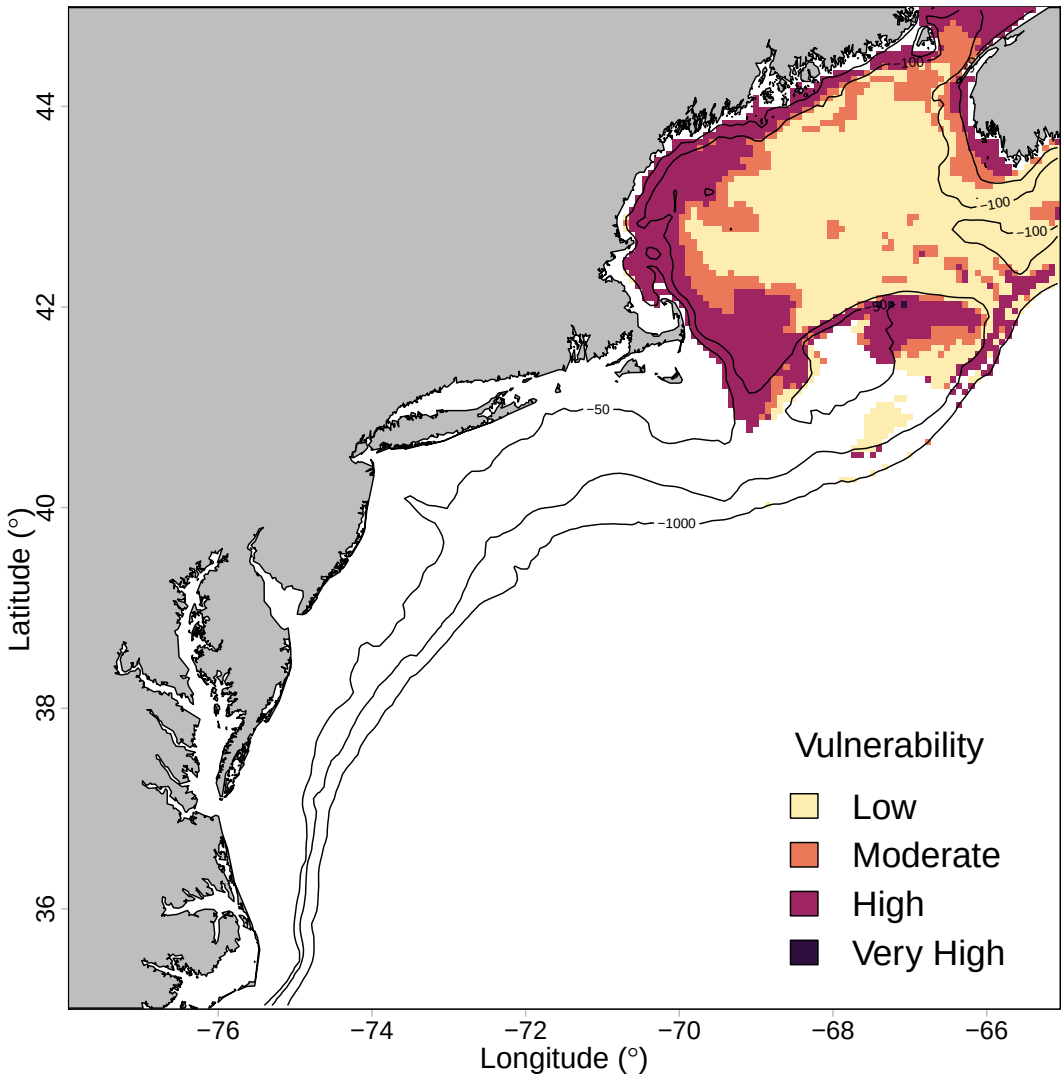


Figure 5: Vulnerability on the Northeast Shelf for Atlantic halibut. Vulnerability is the product of sensitivity and exposure (see Figure 4 and Figure 2)

Mean Vulnerability for Atlantic halibut was **Moderate** (5.47 +/- 2.70) (Figure 1). Vulnerability was High along the coast of the Gulf of Maine and over Georges Bank, but Low throughout the rest of the Gulf (Figure 5). In the previous Assessment, halibut received a High vulnerability score, driven by High exposure and sensitivity (Hare et al. 2016).

Data Quality Metrics

Life History Data Quality:

87% of data quality scores for the sensitivity attributes were greater than or equal to 2, indicating strong data quality. Stock Size/Status was the only identified data gap (mean attribute data quality score <

2).

Model Confidence Score:

Mean model confidence for Atlantic halibut was 1.7 (+/- 0.4), indicating low confidence in the models. Experts highlighted that residuals were noisy because the species is caught infrequently in the surveys used. The ensemble model's AUC was 0.75, indicating fair performance.

Additional Metrics

Distribution Shifts:

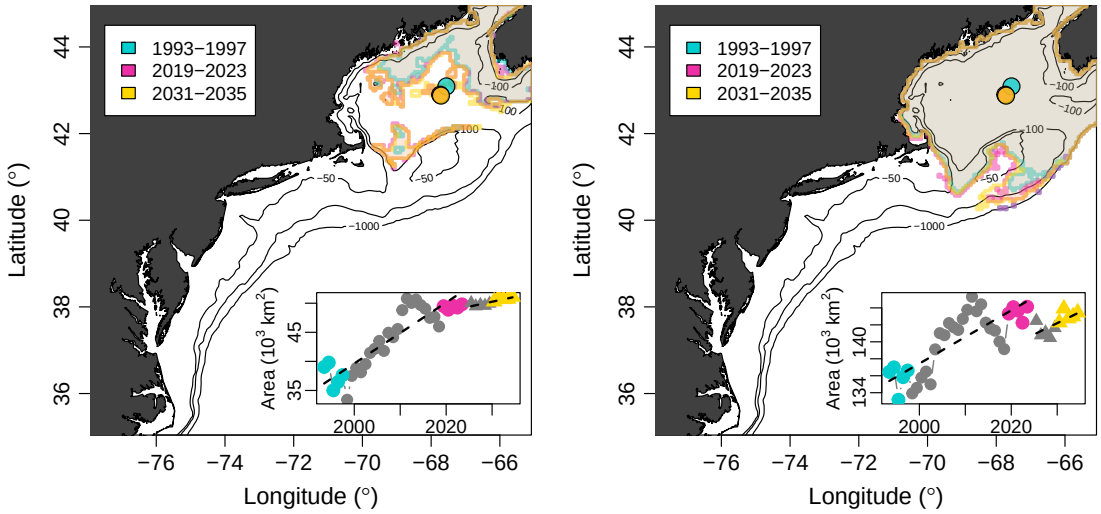


Figure 6: Distribution shifts on the Northeast Shelf for Atlantic halibut. Polygons illustrate core habitat (left) and species range (right) for five-year average periods. Points illustrate weighted centroid, calculated as the average position with species distribution models as weights. Inset timeseries illustrate the shift in core and habitat range area over time. Circles represent points from the MOM6 hindcast; triangles represent points from the MOM6 forecast. Dashed lines represent significant linear regressions

Note: For an animated version of this plot, please see the online version of this document.

Between 1993 and 2023, distribution weighted centroid for halibut shifted southwest 26 km. This was due to habitat expansion at the southern edge of the range (Figure 6). Over time, the probability of finding halibut in the Gulf of Maine increased, and core habitat expanded from the northern Gulf to the northern flank of Georges Bank (Figure 6; **?@fig-movement-halibut**).

Under near future ocean conditions, the distribution centroid is predicted to shift a further 3 km southeast and the southern edge of the range is expected to continue shifting southward. These changes are predicted to significantly increase the available habitat for halibut in the region (Figure 6).

In the previous assessment, halibut were given a High Distributional Vulnerability Rank because adults are habitat generalists and early life stages can disperse far distances (Hare et al. 2016).

Table 2: Distribution shifts based on five-year average distributions. Changes are calculated relative to the previous time period.

Time Period	Weighted Centroid Shift (km & bearing)	Core Habitat Area (change) (km^2)	Habitat Total Area (change) (km^2)	Latitude Range (change) (deg)	Change in Longitude Range (deg)
1993-1997	NA	37,647	135,969	3.55	4.57
2019-2023	25.9 km SW	49,434 (+11,786)	143,620 (+7,651)	3.55 (0)	4.57 (0)
2031-2035	3.0 km SE	50,748 (+1,315)	143,083 (-538)	3.55 (0)	4.65 (0.08)

Directional Effect:

Halibut will likely be **negatively** impacted by climate change (100% certainty in expert scores). While halibut distributions are expanding due to warmer temperatures, they are shifting northward and off the Northeast shelf into Canadian waters. This is similar to the result from the 2016 Assessment (Hare et al. 2016).

Climate Effects on Abundance and Distribution

Nye et al. (2009) observed a northward shift in Atlantic halibut distribution and an increase in the depth of occurrence on the Northeast U.S. continental shelf. The species' thermal habitat has widened, likely contributing to its range expansion (Shackell et al. 2022), and biomass and landings in Canadian waters have thus increased greatly since the early 2000s (Trzcinski and Bowen 2016). Shackell et al. (2022) calculated growth rates from tagging studies and found faster growth in Atlantic halibut in warmer, southern waters. From 1990 to 2018, thermal habitat rose by 11.6% (+/- 7.35%) and the number of degree days where Atlantic halibut are experiencing growth increased by 13.5 °C days (+/- 7.86 °C days) (Czich et al. 2023). Continued warming is likely to increase abundance in the northern end of the species' range; however elevated CO^2 levels and rising temperatures may lead to oxidative stress (Almroth et al. 2019).

Life History Synopsis

Atlantic halibut are found from the Labrador Shelf in the North to Virginia, although they are rare south of Long Island (Cargnelli, Griesbach, and Morse 1999). Adult halibut spawn from late fall to late spring in deep continental shelf waters, with presumed spawning peaks occurring at 2 - 9 day intervals between October and January (Collette and Klein-MacPhee 2002) (Armsworthy, Trzcinski, and Campana 2014). Eggs hatch in 13-20 days depending on the temperature (Blaxter et al. 1983); larvae have been found in the plankton as much as a year after spawning (Collette and Klein-MacPhee 2002). Depending on current speeds and larval duration, eggs and larvae may drift anywhere from 300-500 km. Tagging studies of spawning individuals suggest that larvae can be transported into the Gulf of St Lawrence, which may provide nursery habitats (Armsworthy, Trzcinski, and Campana 2014). Juveniles are most common in shallow water from 20-60 m in nursery areas on the shelf (Cargnelli, Griesbach, and Morse 1999). Juvenile halibut up to 30 cm in length consume invertebrates, mostly annelid worms and crustaceans; they then feed on fish and invertebrates from 30-80 cm in length, and then almost exclusively on fish as they grow larger (Scott and Scott 1988) (Bowman et al. 2000). Adults normally inhabit benthic waters from 2-15°C,

but occupy the entire water column to feed and spawn (Cargnelli, Griesbach, and Morse 1999) (Collette and Klein-MacPhee 2002) (Armsworthy, Trzcinski, and Campana 2014). In the Gulf of Maine, tagged individuals have been observed conducting both shore and long distance migrations, as well as being resident year-round (Shackell et al. 2022). Age at maturity ranges from 7 - 13 years (females) and 5 - 12 years (males), and their maximum age is between 38-50 years (Armsworthy, Trzcinski, and Campana 2014). Atlantic halibut is currently managed by the New England Fishery Management Council under the Northeast Multispecies Fishery Management Plan. The most recent management track assessment was completed in 2024, where stock status could not be determined analytically due to a lack of biological reference points (Northeast Fisheries Science Center (NEFSC) 2024).

Supplemental Scores

Exposure Timeseries

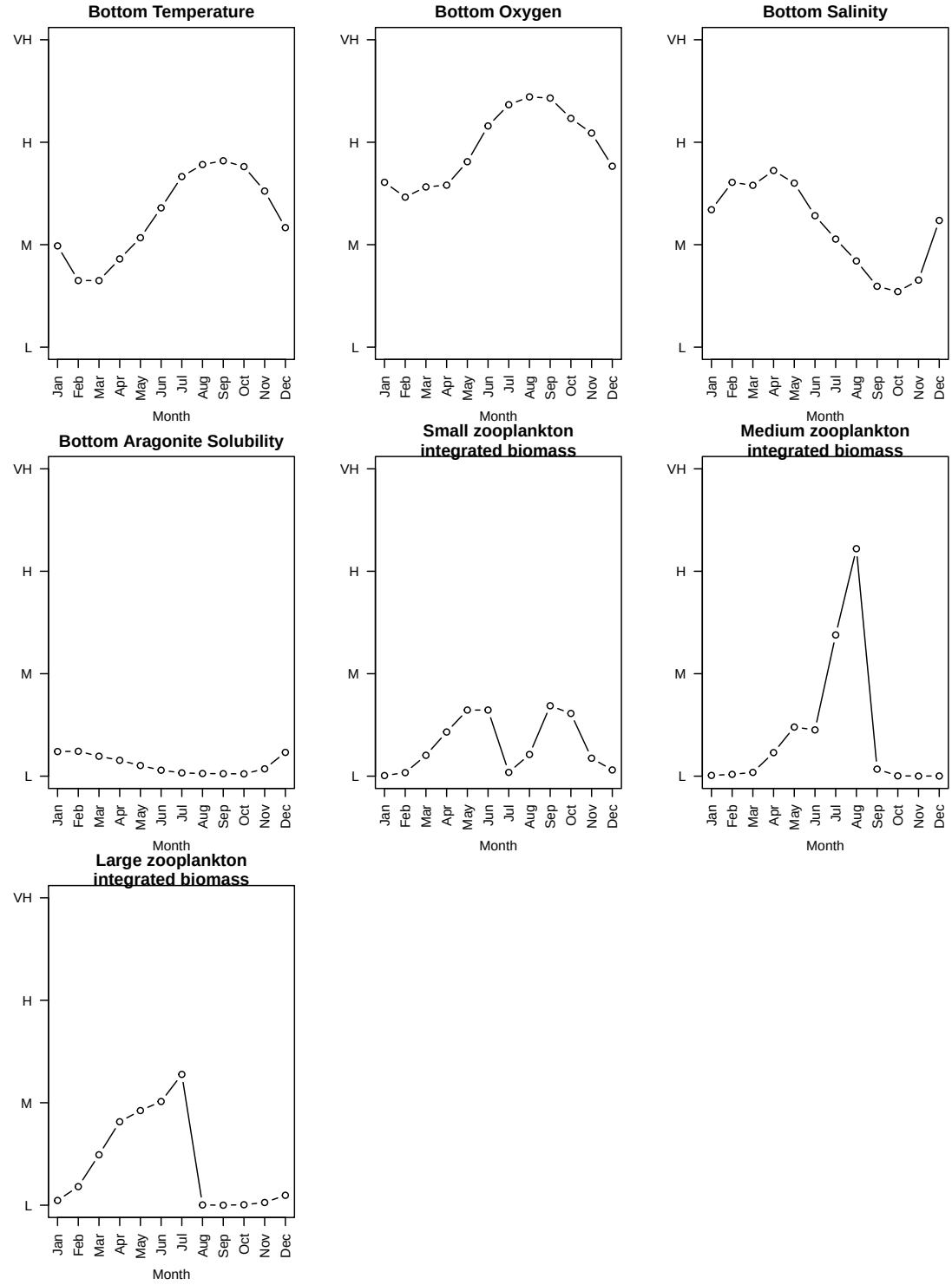


Figure 7: Monthly Timeseries of each Exposure Factor for Atlantic halibut. Exposure was calculated between 2014-2023 and 2025-2035

Across months, exposure ranged from Low to High. High exposure occurred in August, due to spikes in medium zooplankton biomass (Figure 8, Figure 7).

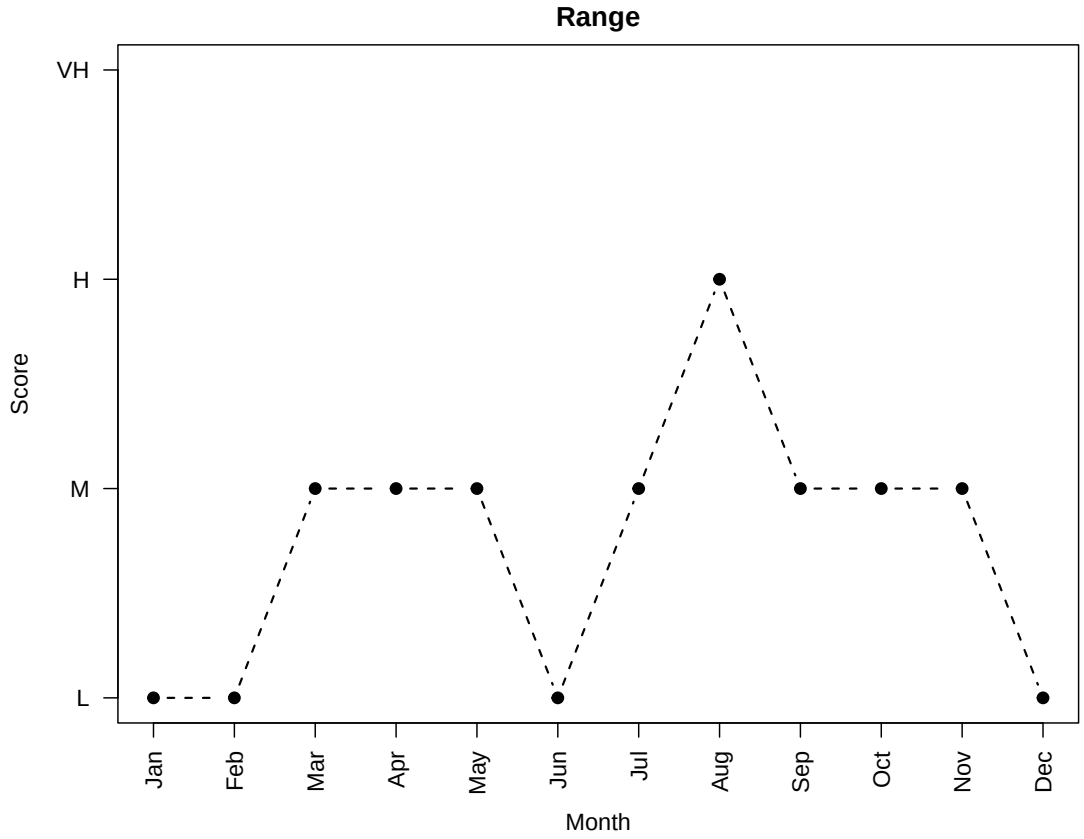


Figure 8: Timeseries of Total Exposure for Atlantic halibut

Exposure due to Significant Factors

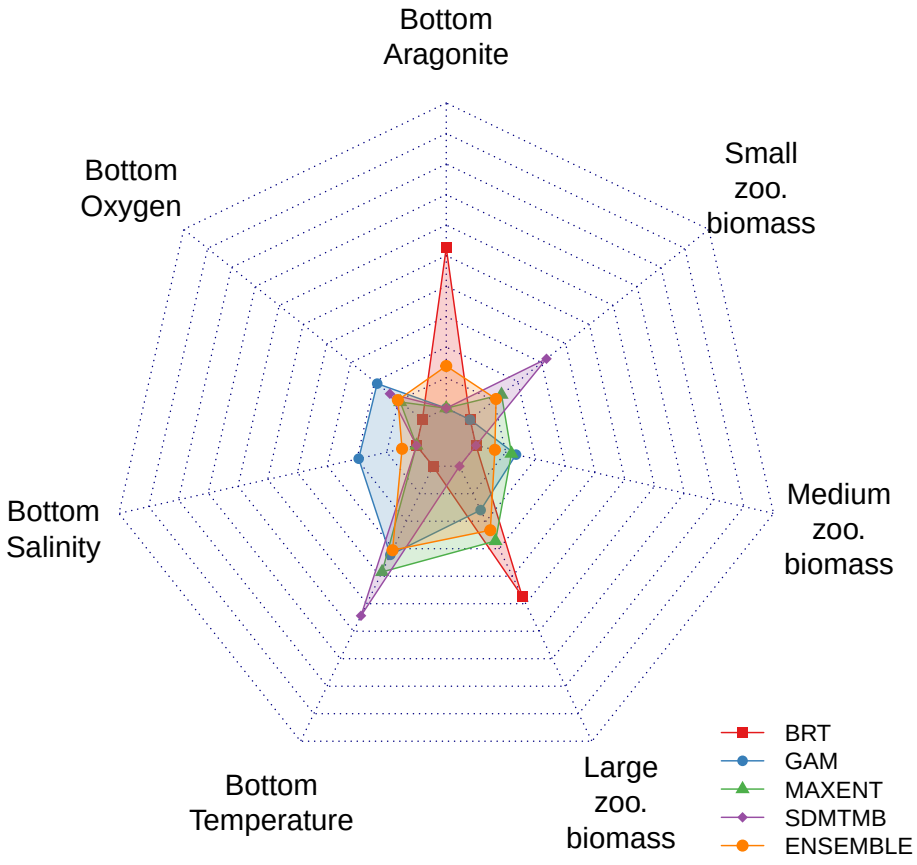


Figure 9: Dynamic Factors driving Atlantic halibut distributions. The closer the polygon is to the edge of the figure, the more important the variable is. Each ring represents 10% variance explained. Factors must explain at least 10% of variance in the ensemble model to be used to calculate total exposure.

Bottom temperature, oxygen, and aragonite solubility as well as small and large zooplankton biomass were important dynamic drivers of halibut distributions (Figure 9). Halibut occupy benthic habitats, and consume fish and invertebrates, including crustaceans (Cargnelli, Griesbach, and Morse 1999; Collette and Klein-MacPhee 2002; Armsworthy, Trzcinski, and Campana 2014; Scott and Scott 1988; Bowman et al. 2000). When Exposure was calculated with only these factors, mean total exposure remained Moderate (1.69 +/- 0.88) (Figure 10; Figure 11).

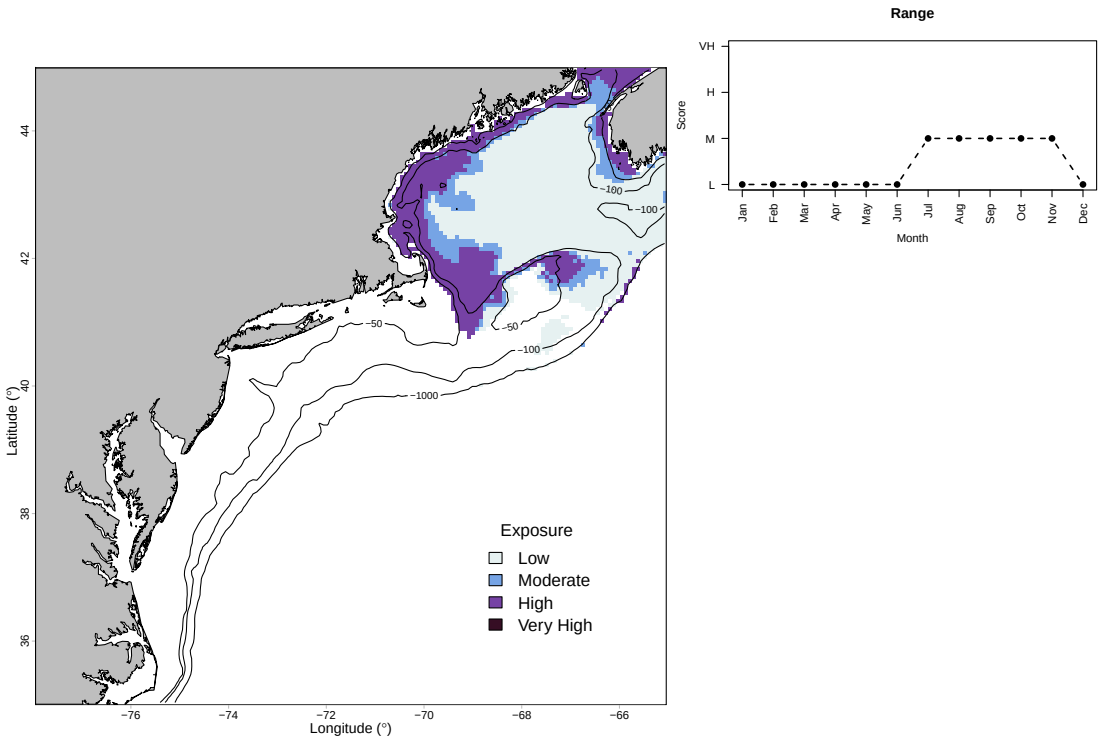
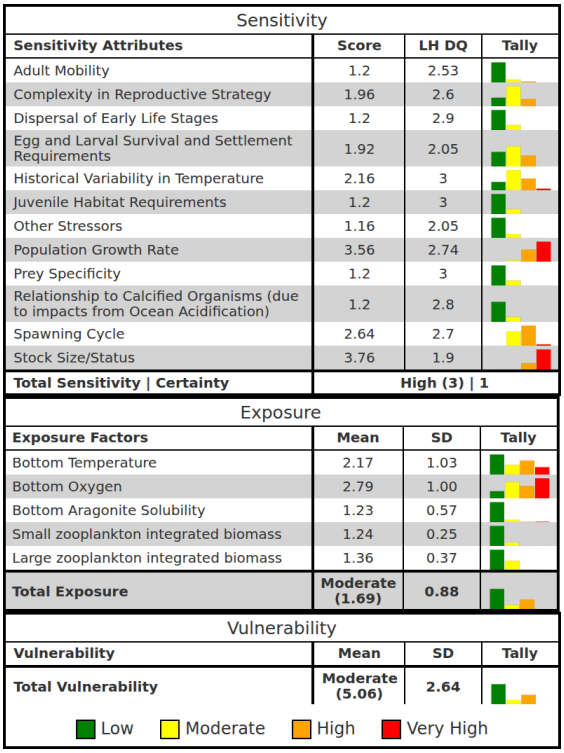


Figure 10: Total Exposure Map and Timeseries when Exposure was calculated with only Exposure Factors critical to Atlantic halibut distributions



Sensitivity			
Sensitivity Attributes	**Score**	**LH DQ**	**Tally**
Adult Mobility	1.2	2.53	
Complexity in Reproductive Strategy	1.96	2.6	
Dispersal of Early Life Stages	1.2	2.9	
Egg and Larval Survival and Settlement Requirements	1.92	2.05	
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Spawning Cycle	2.64	2.7	
Stock Size/Status	3.76	1.9	
Total Sensitivity | Certainty	**High (3) | 1**		

Exposure			
Exposure Factors	**Mean**	**SD**	**Tally**
Bottom Temperature	2.17	1.03	
Bottom Oxygen	2.79	1.00	
Bottom Aragonite Solubility	1.23	0.57	
Small zooplankton integrated biomass	1.24	0.25	
Large zooplankton integrated biomass	1.36	0.37	
Total Exposure	**Moderate (1.69)**	**0.88**	

Vulnerability			
Vulnerability	**Mean**	**SD**	**Tally**
Total Vulnerability	**Moderate (5.06)**	**2.64**	

Low Moderate High Very High

Figure 11: As in Figure 1 for Atlantic halibut, but with only significant Exposure factors and the resulting Vulnerability Scores.

References

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